SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Cyber Forensics & Incident Response: VAPT, Memory Forensics & Laws

Department: CSE CS	**Course / Subject:** Cyber Forensics & Incident Response (CS605PC)
Semester: Semester 6	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Vulnerability Assessment & Penetration Testing (VAPT), Volatile memory acquisition, Disk forensic imaging (dd/E01), Network packet analysis, and Cyber laws (IT Act 2000).

Unit 1: Digital Forensics Fundamentals

Digital evidence types, Locard's exchange principle, Chain of custody, Forensic readiness, and Evidence seizure protocols under ISO 27037.

Unit 2: Disk & File System Forensics

Bit-stream disk imaging, Write blockers, FAT/NTFS file systems, Master File Table (MFT), Deleted file recovery, Slack space analysis, and Steganography detection.

Unit 3: Memory & Network Forensics

Volatile memory capture (RAM dumping), Volatility framework analysis, Malicious process injection detection, Wireshark packet capture, and IDS/IPS log analysis.

Unit 4: Incident Response & Threat Hunting

NIST incident response lifecycle (Preparation, Detection & Analysis, Containment, Eradication, Recovery), MITRE ATT&CK; framework mapping, and SIEM tool monitoring.

Unit 5: Ethical Hacking & Legal Framework

Penetration testing phases (Reconnaissance, Scanning, Exploitation, Maintaining access), OWASP Top 10 vulnerabilities, Indian Information Technology Act 2000 amendments.